IFRS 9 climate provision report

Banco Demo (fictional) · season 2022/23 · as of 18 Jan 2023 (week 33). Expected credit loss with a climate overlay, compared with the bank's own baseline (PD and LGD at origination).

Loans in the book
209
of 240 in the file

Outstanding (EAD)
$30.77M
$30,765,200

Exposure High/Severe
$15.69M
51.0% of EAD

Stage 2 / Stage 3 loans
60 / 5
IFRS 9 staging

ECL, bank baseline
$807k
$806,742

ECL, climate-weighted
$1.45M
$1,445,988

Provision change
+$639,246
+79% vs bank

1-in-20 loss (p95)
$1.35M
historical replay

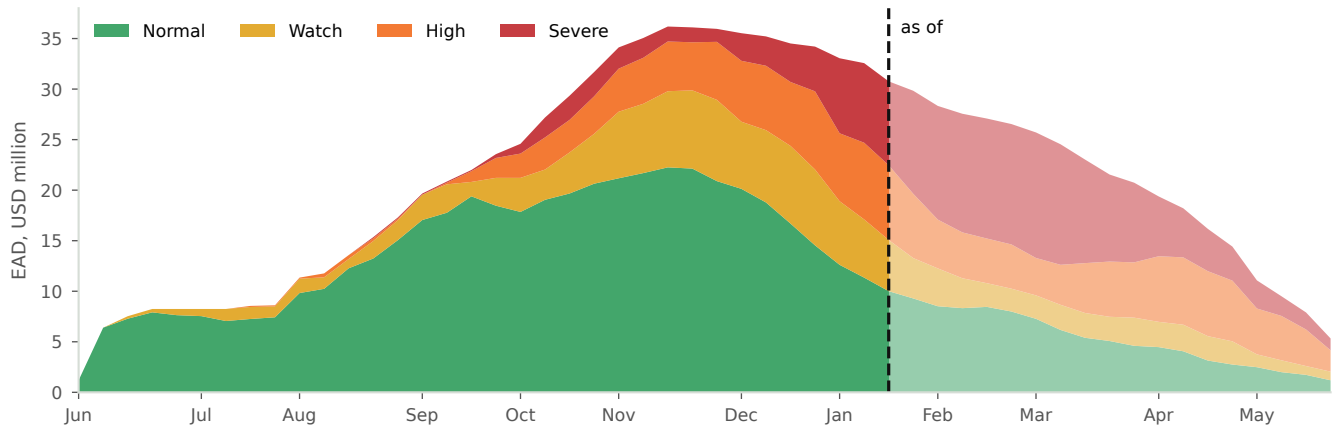
ECL by scenario

Scenario	Weight	ECL (USD)	vs bank
Bank baseline (PD, LGD at origination)	-	$806,742	-
Climate: base	60%	$1,329,072	+65%
Climate: adverse	25%	$1,564,060	+94%
Climate: severe	15%	$1,716,863	+113%
Climate: probability-weighted	**100%**	**$1,445,988**	**+79%**

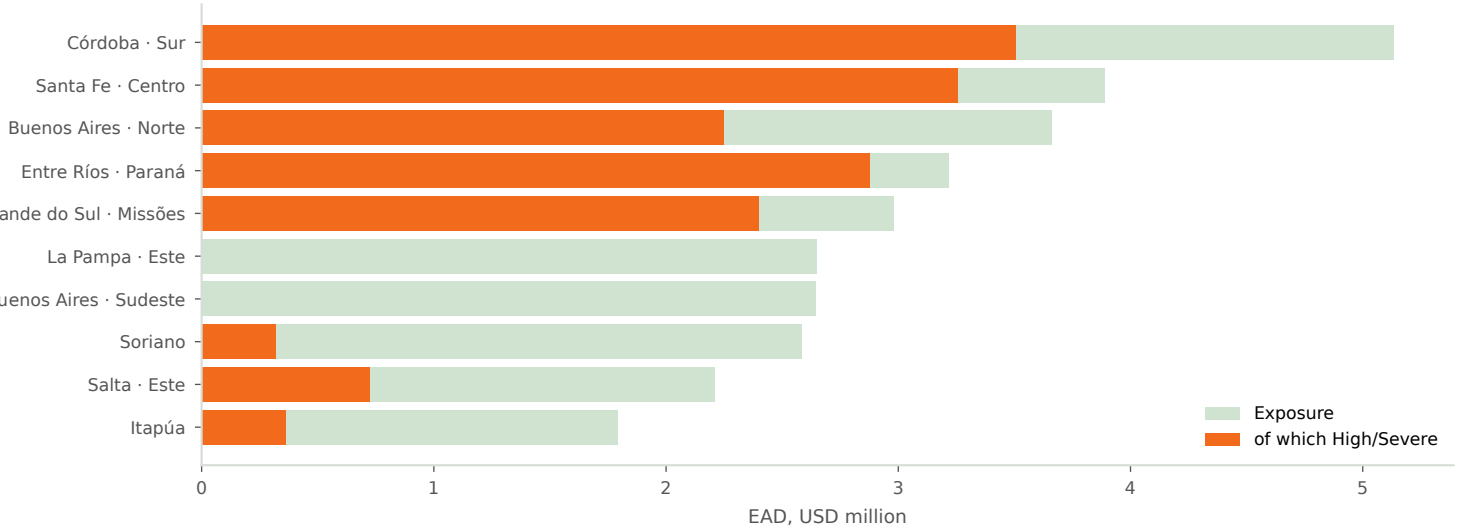
ECL by IFRS 9 stage

Stage	Loans	EAD	ECL bank	Base	Adverse	Severe	Weighted
Stage 1 (12-month ECL)	144	$19,978,300	$244,779	$306,337	$369,282	$493,580	$350,161
Stage 2 (lifetime ECL)	60	$9,765,900	$153,349	$606,983	$777,290	$796,626	$678,009
Stage 3 (credit-impaired)	5	$1,021,000	$408,612	$415,751	$417,484	$426,654	$417,820

Exposure by stress level over the season



Concentration by region



Concentration by crop

Crop	Loans	EAD	High/Severe EAD	Share stressed	Provision change
Soybean	123	$15,989,900	$8,396,900	52.5%	$265,042
Maize	44	$10,357,700	$6,204,400	59.9%	$339,834
Wheat	24	$2,256,400	$961,600	42.6%	$24,172
Sunflower	18	$2,161,200	$130,700	6.0%	$10,202

Clusters of stressed loans (DBSCAN, 60 km)

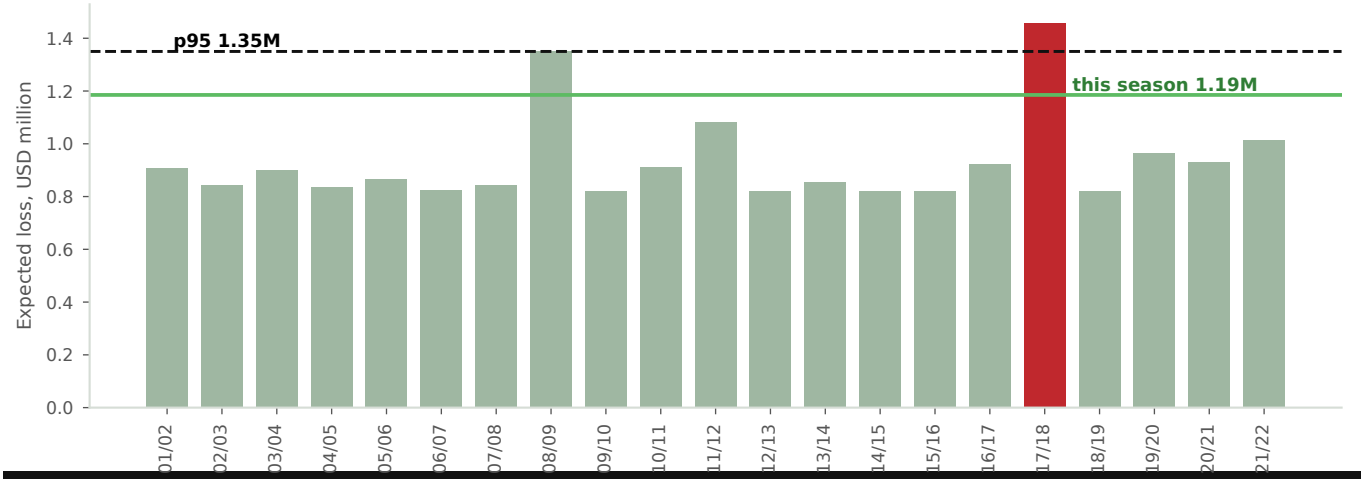
Cluster	Loans	Main region	EAD	Mean score	Radius
#0	39	Santa Fe · Centro +	$5,771,100	72	153 km
#2	14	Córdoba · Sur	$3,228,100	80	99 km
#1	16	Buenos Aires · Norte	$2,247,000	68	108 km
#3	17	Rio Grande do Sul · Missões	$2,202,400	74	80 km

Loans with the largest provision change

Loan	Borrower	Crop	Level	Stage	EAD	Change
L093	Campo Los Cardos S.R.L.	maize	Severe	Stage 2	$875,000	+$152,595
L037	Cooperativa La Victoria S.A.	soy	Severe	Stage 2	$200,000	+$36,048
L090	Establecimiento San José	maize	Severe	Stage 2	$369,000	+$24,801
L117	Estancia El Porvenir S.R.L.	maize	Severe	Stage 2	$239,400	+$22,469
L105	Estancia El Recreo S.A.	maize	Severe	Stage 2	$216,000	+$21,409
L107	Establecimiento El Mirador S.A.S.	maize	Severe	Stage 2	$157,000	+$20,147
L097	Campo Los Talas S.R.L.	maize	Severe	Stage 2	$316,000	+$15,343
L108	Granja Santa Elena	soy	Severe	Stage 2	$89,000	+$15,259

Loan	Borrower	Crop	Level	Stage	EAD	Change
L004	Campo Santa Elena S.A.	soy	**Severe**	Stage 2	$105,000	+$14,143
L072	Campo La Rinconada	maize	**Severe**	Stage 2	$165,000	+$12,549

Stress test: today's book replayed through past seasons



Measure	Expected loss (USD)	% of EAD
Median season (p50)	$865,108	2.8%
1-in-20 years (p95)	$1,349,979	4.4%
99th percentile	$1,434,643	4.7%
Worst season (2017/18)	$1,455,809	4.7%
This season (base scenario)	$1,185,171	3.9%

Expected loss (PD x LGD x EAD) of today's book if each history season repeated; Stage 3 loans included in every year.

Methods and caveats

- anomalies: Per-field weekly climatology over 21 seasons (mean, sd, p10, p90); z-score and percentile rank; SPI-30/60/90 from a gamma fit with a zero-rain probability; soil moisture percentile.
- stress: Rule-based perils (drought cascade, excess water, frost, heat, hail); intensity x stage weight accumulated with exponential decay (half-life 4 weeks) into a 0-100 score; levels with hysteresis (up 25/45/70, down 20/38/62); confidence = indicator agreement x location quality x cloud-free share.
- yield impact: Per region x crop: OLS (p50) and quantile regression (p10, p90) of department yield anomaly on season stress indicators; leave-one-year-out validation; weekly prediction from indicators available to date, wider before sowing.
- ifrs9: logit(PD) = logit(PD_orig) + beta x yield loss; Stage 2 on a critical alert or PD >= 2x origination, Stage 3 on BCRA situacion >= 3 or > 90 days past due; LGD x (1 - insurance cover share) with collateral adjustment; EAD from the amortization schedule; ECL = sum of marginal PD x LGD x EAD x DF monthly (12 months for Stage 1, lifetime for Stage 2/3; Stage 2 on short crop loans adds one year of assumed restructuring term); scenarios base/adverse/severe 60/25/15. IFRS 9-aligned, not audit-ready: PD adjustments are assumptions to calibrate on the lender's own data.
- portfolio: Weekly totals by level and stage; DBSCAN (haversine, 60 km, min 3) clusters of High/Severe loans; concentration by region and crop; historical-simulation stress test of today's book through the 21 history seasons (p95 = 1-in-20-year loss).
- alerts: Alerts from hysteresis-filtered level transitions; template explanations from computed numbers (optional LLM rewording, template fallback); rule-table recommendations.

Parameters: beta 3.0 (logit PD per unit of yield loss), SICR multiplier 2.0x, discount rate 8%, scenario weights base 60%, adverse 25%, severe 15%, hysteresis up [25.0, 45.0, 70.0] down [20.0, 38.0, 62.0].

Caution: this is IFRS 9-aligned, not audit-ready. The beta, the SICR multiplier, the scenario weights, the PD-by-rating and LGD-by-collateral tables and the yield-to-PD link are assumptions until calibrated on the lender's own repayment history. All satellite, weather, yield and loan data in this demo are synthetic.